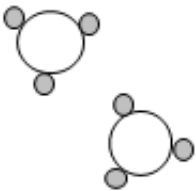


| Question |     |       |  | Marking details                                                                                                                                                                                                                                                               | Marks available |     |     |       |       |      |
|----------|-----|-------|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |       |  |                                                                                                                                                                                                                                                                               | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 9/2      | (a) | (i)   |  | air<br><br>do not accept oxygen                                                                                                                                                                                                                                               | 1               |     |     | 1     |       |      |
|          |     | (ii)  |  | award (1) for any sensible answer e.g.<br><br>strong(er) equipment required<br>requires thick(er) pipes<br>requires strong(er) pipes<br><u>more</u> maintenance<br><u>may</u> explode<br><u>more</u> energy needed<br><u>more</u> expensive<br><br>neutral answer - dangerous | 1               |     |     | 1     |       |      |
|          |     | (iii) |  | catalyst                                                                                                                                                                                                                                                                      | 1               |     |     | 1     |       |      |
|          |     | (iv)  |  | <br><br>award (2) for correct answer<br>award (1) for one ammonia molecule drawn correctly<br><br>award (1) max if any additional product(s) included                                       |                 | 2   |     | 2     |       |      |

| Question |     |  |  | Marking details                                                                 | Marks available |     |     |       |       |      |
|----------|-----|--|--|---------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |  |  |                                                                                 | AO1             | AO2 | AO3 | Total | Maths | Prac |
|          | (b) |  |  | A                                                                               |                 |     | 1   | 1     |       |      |
|          | (c) |  |  | $2\text{NH}_3 + 3\text{Cl}_2 \longrightarrow \text{N}_2 + \boxed{6} \text{HCl}$ |                 | 1   |     | 1     |       |      |
|          |     |  |  | Question 9/2 total                                                              | 3               | 3   | 1   | 7     | 0     | 0    |